CS & IT ENGINEERING



Algorithms

Analysis of Algorithms



Lecture No.- 04

By- Aditya sir

Recap of Previous Lecture



Topic — Best Case, Worst Case

Topic — Asymptotic Notations

O, Ω, Θ

Topics to be Covered



- Topic
- Topic
- Topic

Practice Questions
+
PYQs
+
Small Notations



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Asymptotic Notations

(2) Big Omega (Ω):

$f(n)$ & $g(n)$

$f(n) = \Omega(g(n))$

iff, $f(n) \geq c * g(n)$; $n > n_0$, $c > 0$

e.g. $7n^3 + 8n + 1000 \Rightarrow \Omega(n^3)$

↳ True

Big Oh

$f(n) = O(g(n))$ iff

$f(n) \leq c * g(n)$,

$c > 0$

$n \geq n_0$



Topic : Asymptotic Notations

(3) Theta (θ):

Tight Bound

$f(n)$ & $g(n) \Rightarrow$ $f(n) = \theta(g(n))$

iff, $f(n) = O(g(n))$

$f(n) = \Omega(g(n))$

or $c_2 * g(n) \leq f(n) \leq c_1 * g(n)$

Hence, $f(n) = \theta(g(n))$

e.g.: $9n^3 + 7n^2 + 100$

$\Rightarrow f(n) = O(n^3)$

$= \Omega(n^3)$

$> \theta(n^3)$

eg:-

$$f(n) = 100 * \log_2 n + 50\sqrt{n}$$

Appx1 :-

$$\log_2 n < \sqrt{n}$$

n=64

$$\log_2(64) \qquad \sqrt{64}$$

$$\downarrow \qquad \downarrow$$

$$6 < 8$$

Appr2 : take log( )

n^3

$\log(n^3)$

$(3 \log n)$

$\log n < \sqrt{n}$

$\log n$ $\quad$ $(n)^{1/2}$

Take log both sides

$\log(\log n)$ $\quad$ $\log(n^{1/2})$

$\log \log n$ $\quad$ $\frac{1}{2} * \log n$

Let $x = \log n$

$\log(x) < \frac{x}{2}$

Possible mistake :

n^3 n^5

Take log

$\log(n^3)$ $\log(n^5)$

$3 \log n$ $5 \log n$

$3 < 5$ → maths

$O(1)$ = $O(1)$

→ asymt X



Topic : Asymptotic Notations

Important observations:

(1) ~~Longer~~ larger functions are always omega of the smaller functions.

$2^n = \Omega(n^2)$

$2^n = \Omega(n^3)$

$2^n \geq c * n^2$

e.g.: $n^3 \geq c * n^2$

$n^3 = \Omega(n^2)$



Topic : Asymptotic Notations

Important observations:

(2) Smaller functions are always order (Big Oh) of the larger functions.

$n^2 = O(2^n)$

$n^3 = O(2^n)$

e.g.: $n^3 \geq c * n^2$

$n^2 = O(n^3)$

$n^2 = O(n^3)$

$n^2 \leq c * n^3$



Topic : Asymptotic Notations

Important observations:

(3) If two functions have equal rate of growth then they are theta of each other.

$f(n) = 8n^2$

$g(n) = 5n^2$

e.g.: $f(n) = \theta(g(n))$

or

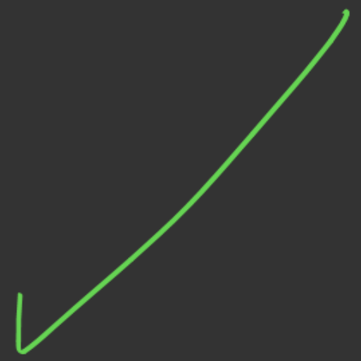
$g(n) = \theta(f(n))$

Quadratic

$f(n) = O(g(n))$

$g(n) = O(f(n))$?

Doubts :





Topic : Exponentials

For all real $a > 0$, m, n

$a^0 = 1$

$a^1 = a$

$a^{-1} = 1/a$

$a^{-1} = \frac{1}{a}$

$a^{-2} = \frac{1}{a^2} = \left(\frac{1}{a}\right)^2$

$a^{-m} = \frac{1}{a^m}$

$(a^m)^n = a^{mn}$

$(a^m)^n = (a^n)^m$

$a^m \cdot a^n = a^{m+n}$



$$(a^m)^n = (a^n)^m = a^{m*n}$$

$$(2^2)^3 = (2^3)^2 = 2^{3\times 2}$$

$$(4)^3 \qquad (8)^2 \qquad 2^6$$

$$64 \qquad 64 \qquad 64$$



Topic : Exponentials

$a^m \times a^n = a^{(m+n)} \neq a^{(m \times n)}$

e.g.:	$2^3 \times 2^2$	$= 2^{(3+2)}$
	8×4	$= 2^5$
	32	$= 32$

$a^m \cdot a^n \neq a^{m+n}$

$a^m \cdot a^n = a^{(m+n)}$

Speed ≈ accuracy

65

77.33/100

neg marking ∅

−0.66

−150



Topic : Analysis of Algorithms

Logarithmic Properties:

$\log X^y = y \log x$

$\log xy = \log x + \log y$

$\log \log n = \log (\log n)$

$a^{\log_b^x} = x^{\log_b^a}$

$a = b\log_b^a$

$\log_b a^n = n.\log_b a$

$\log_b a = \frac{1}{\log_a b}$

$\log n = \log_{10}^n$

$\log^k n = (\log)^k$

$\log \frac{x}{y} = \log x - \log y$

$\log_b^x = \frac{\log_a^x}{\log_a^b}$

$\log_c(ab) = \log_c a + \log_c b$

$\log_b^{1/a} = -\log_b a$

$a^{\log_b c} = c^{\log_b a}$



Topic : Asymptotic Notations

Example: $\log_2(16)$

$\Rightarrow \log_2(4 \times 4) = \log_2 4 + \log_2 4 = 2 + 2 = 4$

$\Rightarrow \log_2(2^4) = \log_2 2^2 + \log_2 2^2 = 4$

$4 \log_2(2)$

$= 4 \times 1$

$= 4$

$$\log_b(a) = \frac{\log_c a}{\log_c b} = \frac{1}{\frac{\log_c b}{\log_c a}}$$

$$= \frac{1}{\log_a b}$$

$$\frac{1}{2/3} = 3/2$$



Topic : Asymptotic Notations

Logarithmic Properties:

$$\log\left(\frac{a}{b}\right) = \log(a) - \log(b)$$

$$\log(a * b) = \log(a) + \log(b)$$

$$\log_b(a) = \frac{\log_c(a)}{\log_c(b)} = \frac{1}{\frac{\log_c(b)}{\log_c(a)}} = \frac{1}{\log_a(b)}$$



Topic : Asymptotic Notations

Logarithmic Properties:

- $\log^2(n) = [\log(n)]^2 = (\log n) * (\log n)$

- $\log_b\left(\frac{1}{a}\right) = \log_b(1) - \log_b(a)$

$= 0 - \log_b(a)$

$= -\log_b a$

$\log_b(1/a) = \log_b(a^{-1}) = -\log_b a$

$\log^2 n \neq \log(n^2)$

$\hookrightarrow (\log n)^2 \neq \log(\log n)$



Topic : Asymptotic Notations

Geometric Progression:

$S = 2^1 + 2^2 + 2^3 + 2^4 = 2 + 4 + 8 + 16 = 10 + 20 = 30$

Or, $r = 2$ and $n = 4$

$$\text{Sum} = \frac{a(r^n - 1)}{(r - 1)} = \frac{2(2^4 - 1)}{(2 - 1)} = \frac{2(16 - 1)}{1} = 30$$

$r > 1$



Topic : Geometric Sum Formula

1. The geometric sum formula for finite terms is given as:

if r = 1, $S_n = n*a$

if |r| < 1, $S_n = \frac{a(1-r^n)}{r-1}$ $S_n = \frac{a(1-r^n)}{1-r}$

if |r| > 1, $S_n = \frac{a(r^n-1)}{r-1}$

Where

- a is the first term
- r is the common ratio
- n is the number of terms

$r = 1$

$$S = 2 + 2 + 2 + 2$$

$$= 2 \times n$$

$$= 2 \times 4 = 8$$



Topic : Asymptotic Notations

Example:

$$\sum_{i=1}^{n} \frac{1}{3^i} = \frac{1}{3^1} + \frac{1}{3^2} + \cdots + \frac{1}{3^n}$$

$r < 1$

n terms, $a = 1/3^1 = 1/3$, $r = 1/3$

$$S_n = \frac{a(1 - r^n)}{(1 - r)} = \frac{\frac{1}{3}\left(1 - \frac{1}{3}^n\right)}{1 - \frac{1}{3}} = \frac{1}{2}\left(1 - \frac{1}{3^n}\right)$$



Topic : Geometric Sum Formula

2. The geometric sum formula of infinite terms is given as:

if $|r| < 1$ $\quad S_\infty = \frac{a}{1-r}$

if $|r| > 1$, the series does not converge and it has no sum.

1) $\sum_{i=1}^{n} i = 1+2+3 \cdots n = \frac{n(n+1)}{2}$

2) $\sum_{i=1}^{n} i^2 = 1^2+2^2+3^2 \cdots n^2 = \frac{n(n+1)(2n+1)}{6}$

3) $\sum_{i=1}^{n} i^3 = 1^3+2^3+ \cdots +n^3 = \left[\frac{n(n+1)}{2}\right]^2$



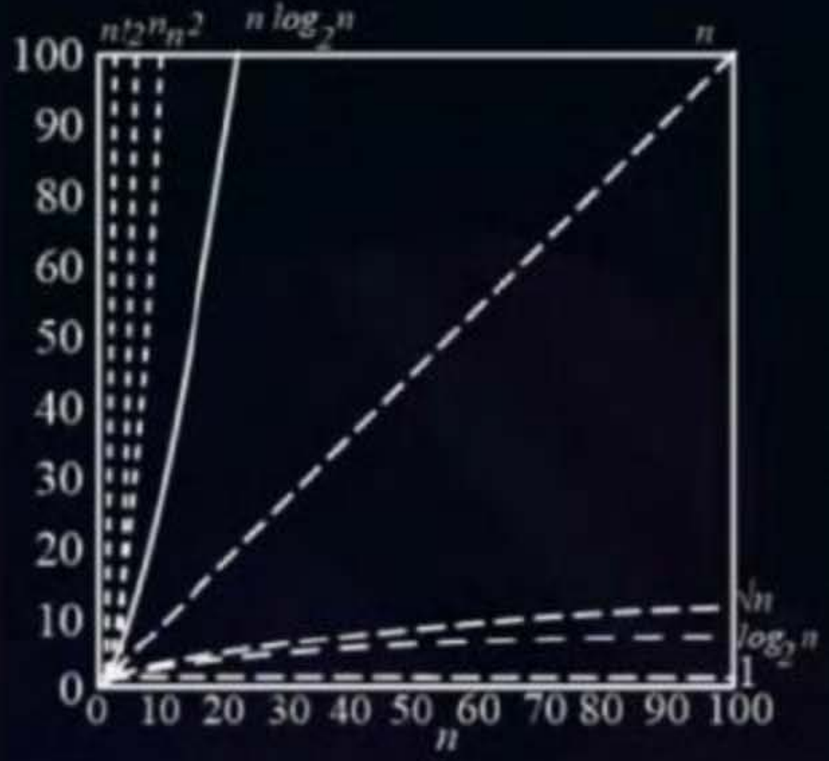
Topic : Analysis of Algorithms

Dominance Relation

Constant < logarithms < poly < exponential

$1 < \log n < \sqrt{n} < n$

$< n^2 < n^3 < n^n$



Practice :-

1) $g(n) = \sum_{b=1}^{n} b$

2) $g(n) = \sum_{b=1}^{n} b^2$

3) $g(n) = \sum_{b=1}^{n} b^3$

4) $g(n) = \sum_{b=1}^{n} 1$

5) $g(n) = \sum_{b=1}^{n} (4)^b$

6) $g(n) = \sum_{b=1}^{n} \left(\frac{1}{6}\right)^b$

7) $g(n) = \sum_{b=1}^{n} n$

Big(Oh)

Big Omega

1) $g(n) = \sum_{b=1}^{n} b = \frac{n(n+1)}{2} = \frac{n^2+n}{2} = \frac{n^2}{2} + \frac{n}{2}$

$O(n^2)$

$\Omega(n^2)$

2) $g(n) = \sum_{b=1}^{n} b^2 = \frac{n(n+1)(2n+1)}{6} = \frac{(n^2+n)(2n+1)}{6}$



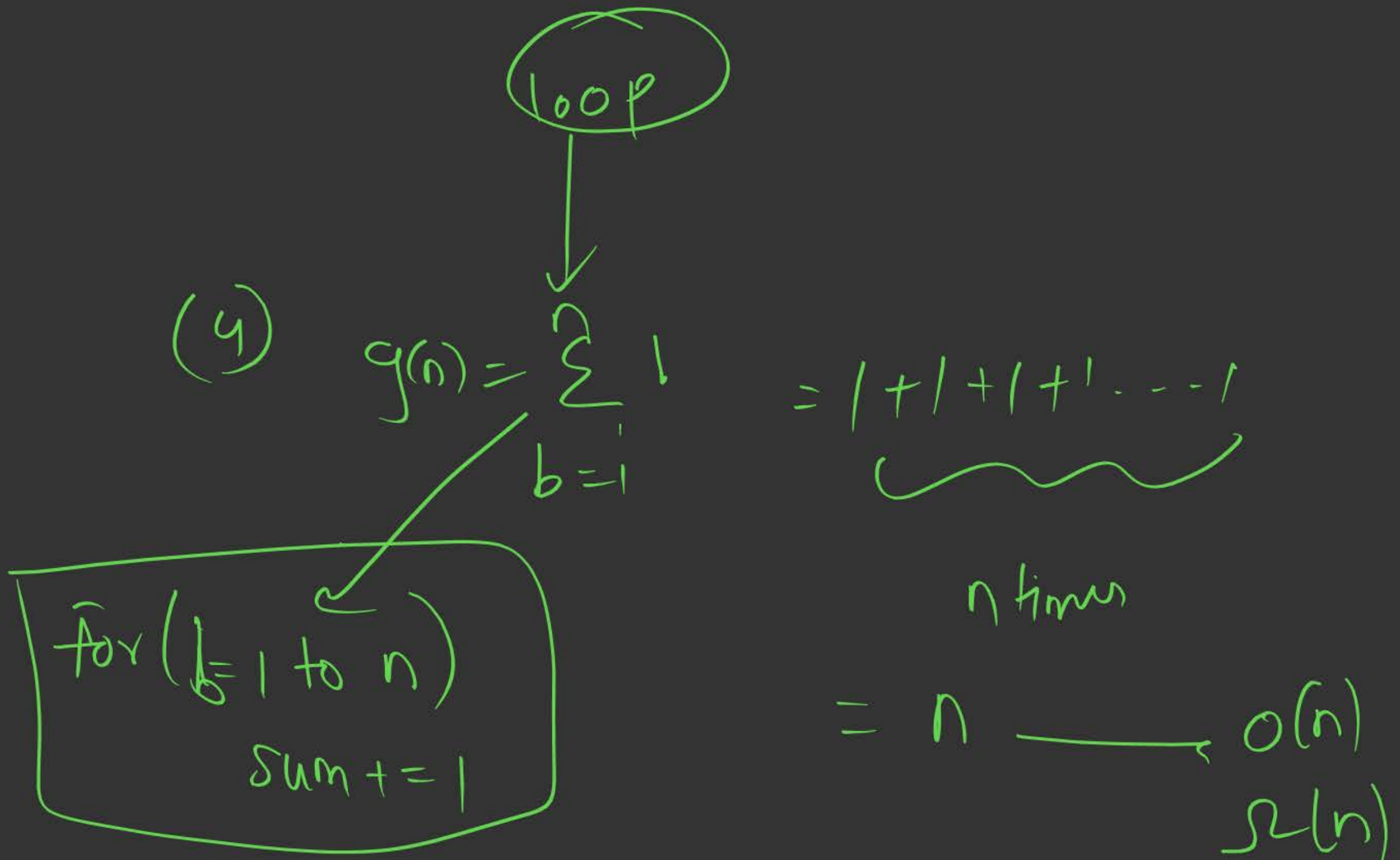
$= \frac{2n^3 + n^2 + 2n^2 + n}{6} = \frac{2n^3 + 3n^2 + n}{6}$

$O(n^3)$

3) $g(n) = \sum_{b=1}^{n} b^3 = \left[\frac{n(n+1)}{2}\right]^2$

$= \left[\frac{(n^2+n)}{2}\right]^2 \approx O(n^4)$

$\Omega(n^4)$


loop
(4) $g(n) = \sum_{b=1}^{n} 1$
$= 1+1+1+1 \cdots 1$
n times
$= n \longrightarrow O(n)$
$\Omega(n)$
for (b = 1 to n)
sum += 1

(5) $g(n) = \sum_{b=1}^{n} 4^b = 4^1 + 4^2 + 4^3 \ldots 4^n$

$\Rightarrow a = 4,\ r = 4,\ n = n$

$g(n) = \frac{a(r^n - 1)}{r - 1} = \frac{4 \times (4^n - 1)}{4 - 1} \rightarrow O(4^n)$

$\Omega(4^n)$

$\frac{4(4^n - 1)}{3} = \frac{4^{n+1} - 4}{3} \longrightarrow O(4^{n+1})$

$\rightarrow O(4 \times 4^n)$

(6) $g(n) = \sum_{b=1}^{n} \left(\frac{1}{6}\right)^b$

$= \left(\frac{1}{6}\right)^1 + \left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^3 \cdots + \left(\frac{1}{6}\right)^n$

GP: $a = 1/6$, $r = \frac{1}{6}$, $n = n$ $\quad r < 1$

$$g(n) = \frac{a(1-r^n)}{1-r} = \frac{\frac{1}{6}\left(1-\left(\frac{1}{6}\right)^n\right)}{1-\frac{1}{6}} = \boxed{\frac{1}{5}\left(1-\frac{1}{6^n}\right)}$$

$$\frac{1}{5}\left(1-\frac{1}{6^n}\right) \neq O\left(\frac{1}{6^n}\right)$$

$1 \geq \frac{1}{6^n}$ → dominating

← Const

→ decr

$$\frac{1}{6} > \frac{1}{6^2} > \frac{1}{6^3}$$

$= O(1)$



(7) $g(n) = \sum_{b=1}^{n} n = n + n + n - - - - + n$

$= n \times n$

$= n^2$

$\Omega(n^2)$

$O(n^2)$ ✓

MSQ

(Q) Given $f(n) = \sum_{i=1}^{n} i^3$ then $F(n)$ is ?

A) $O(n^4)$

B) $O(n^5)$

C) $\Theta(n^5)$ ✗

D) $\Omega(n^3)$

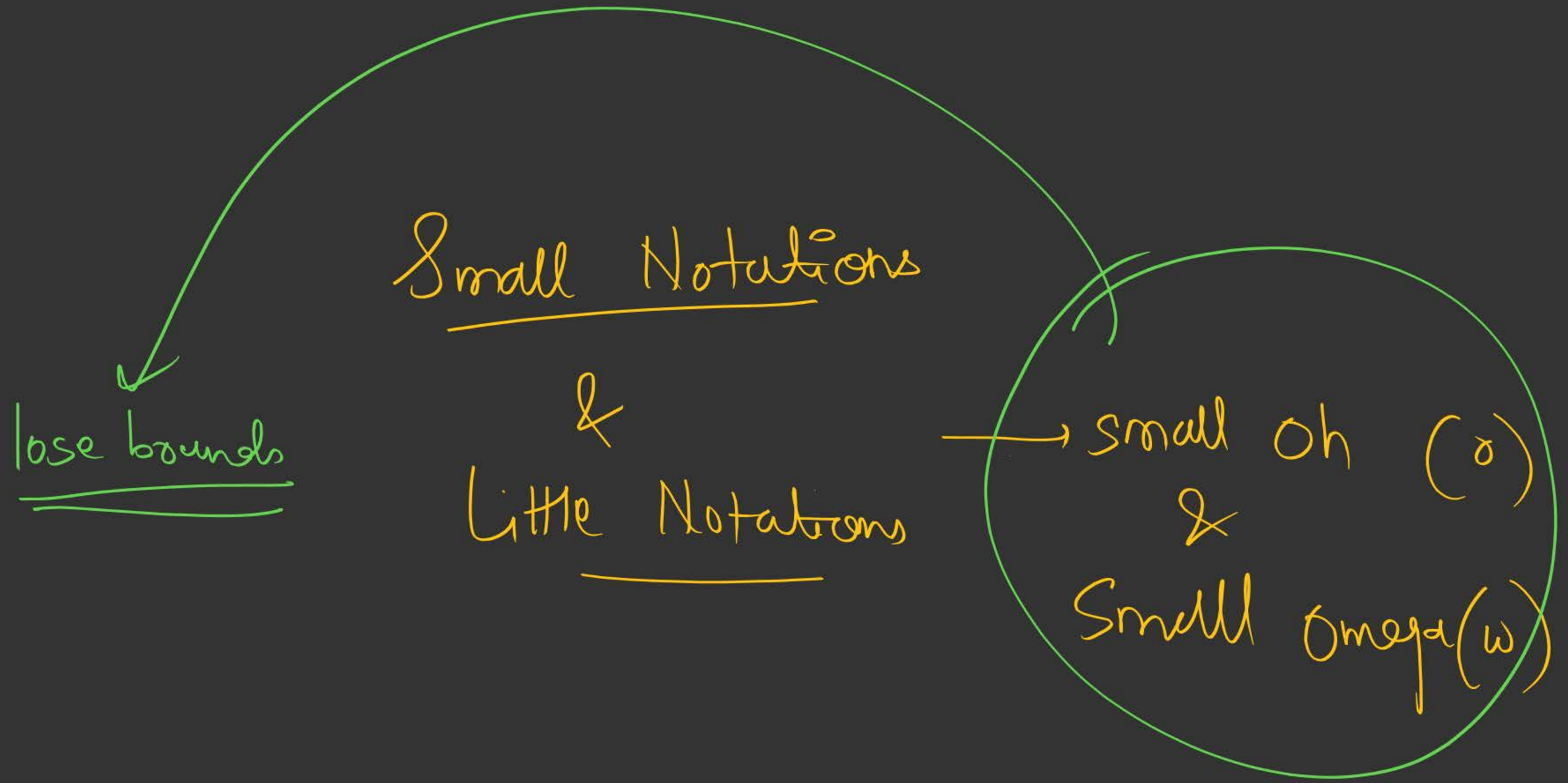
A B D

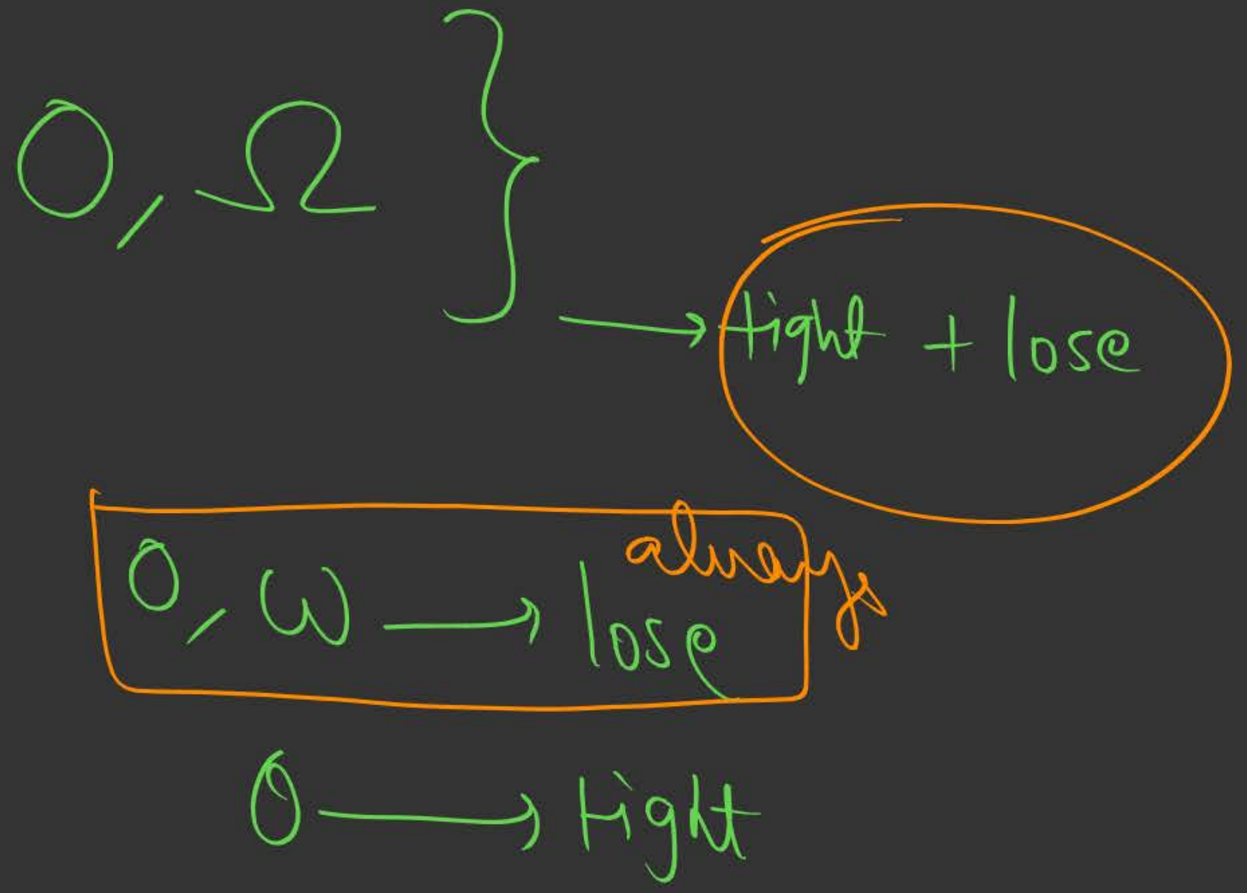
$$\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2 = \left(\frac{n^2+n}{2}\right)^2 = \frac{n^4+2n^3+n^2}{4}$$

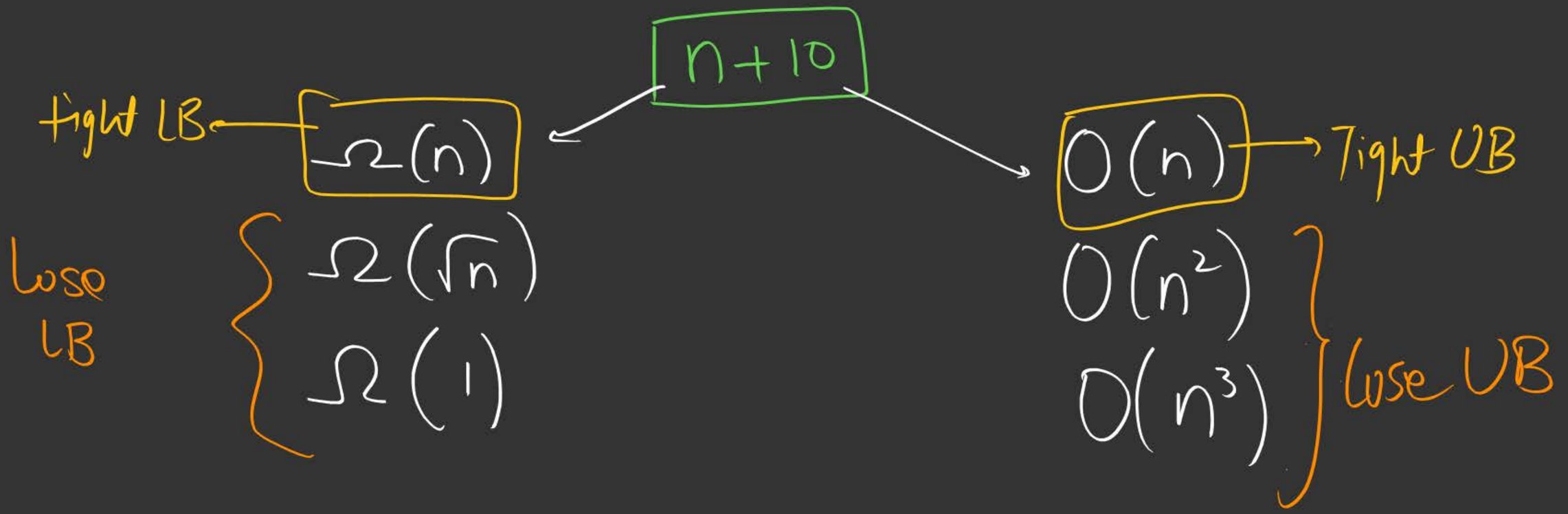
$O(n^4)$ → $O(n^5)$

$\Omega(n^3)$ ← $\Omega(n^4)$

$\Omega(n^5)$


Small Notations
&
Little Notations
lose bounds
small oh (o)
&
Small omega(ω)


0, Ω }
right + lose
0, ω → lose always
0 → right


n+10
tight LB
Ω(n)
Ω(√n)
Ω(1)
Lose LB
O(n)
Tight UB
O(n²)
O(n³)
Lose UB

1) Small Oh (o) → proper UB (loose UB)

$f(n) = o(g(n))$ iff $f(n) < c * g(n)$, for all $c > 0$

when $n \geq n_0$

$f(n) = 2n^2 + 3n$

$\downarrow$

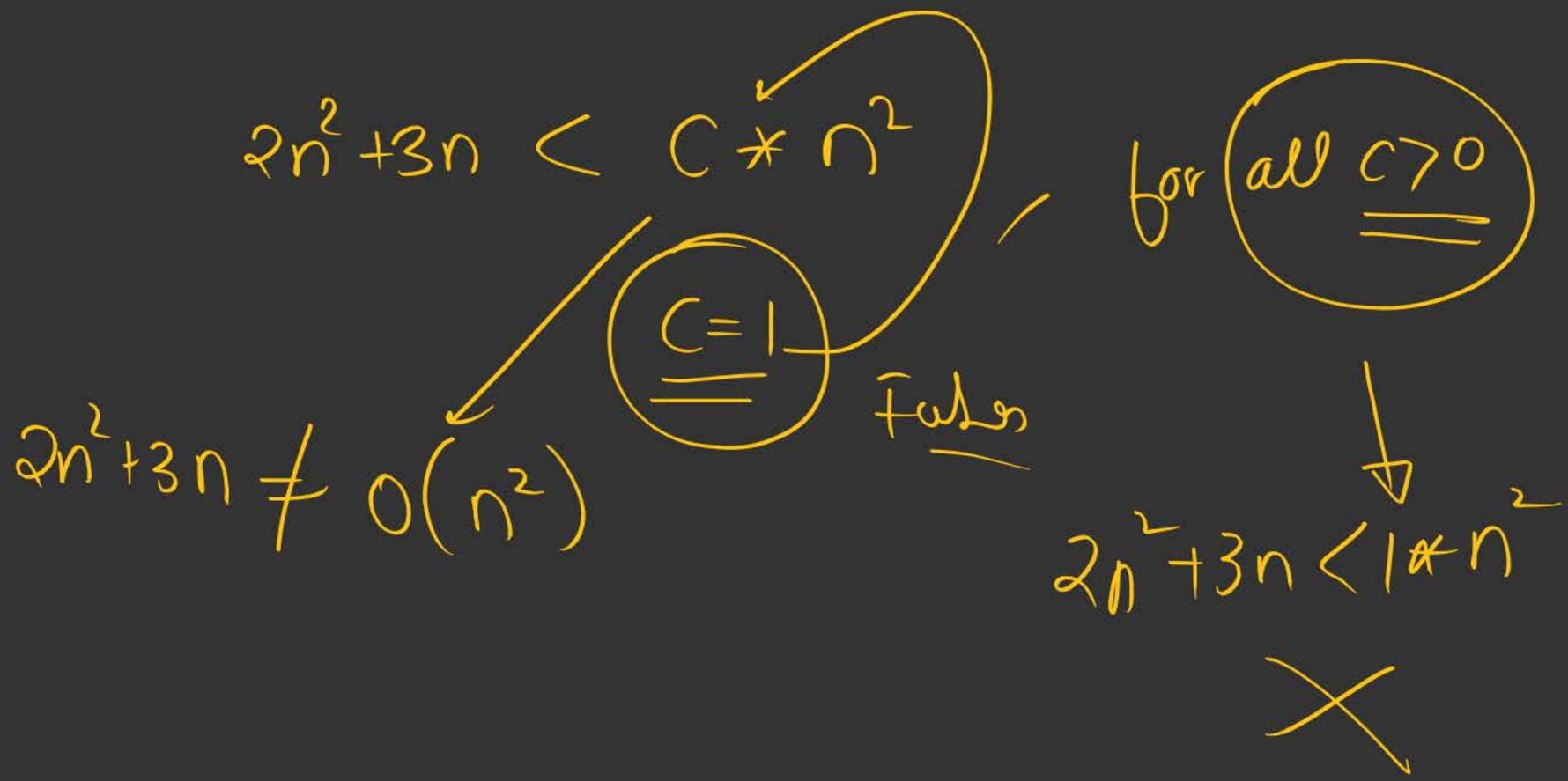
$2n^2 + 3n \leq 10 * n^2$ → Some $C > 0$

$f(n) \leq C * g(n)$

$2n^2 + 3n = O(n^2)$ → Tight UB

→ True

$2n^2 + 3n = O(n^2)$? → False


$2n^2+3n < c*n^2$
for all $c>0$
$c=1$
fails
$2n^2+3n \neq O(n^2)$
$2n^2+3n < 1*n^2$

PYQ

H.W

$W(n)$, $A(n)$ i/p n.

$W(n)$ → Worst Case

$A(n)$ → Avg Case

which is always False

A) $A(n) = o(W(n))$ small oh

B) $A(n) = \Theta(W(n))$

C) $A(n) = \Omega(W(n))$

D) $W(n) = O(A(n))$ small oh



Summary

1) Practice Questns
2) Maths Properties
3) Big, Small



THANK - YOU